Statistical Mechanics: Weak Interactions, Virial Corrections, and Inexact Differentials

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Chapter 1

Weak Interactions, Virial Corrections, and Inexact Differentials

Overview. We begin with the operational meaning of measurement and temperature, then ask how a dilute gas departs from the ideal-gas law when short-range interactions are weakly turned on. The partition function gives the first interaction corrections to energy and pressure and identifies their range of validity. A final mathematical interlude distinguishes state functions from path-dependent heat and work by means of exact differentials, closed loops, and a concrete ideal-gas test.

1.1 What an Instrument Actually Measures

No laboratory device reaches directly into nature and reports an abstract mathematical variable. A spring balance reports an extension; a thermometer reports the height of a liquid column. We call the first a force measurement and the second a temperature measurement only because a reproducible relation has been established between the visible response and the desired quantity.

That relation has a limited range. A spring obeys Hooke's law only while it remains elastic, and a sufficiently large force eventually deforms or breaks it. More generally, if a smooth readout R depends on a physical variable X, then near a chosen operating point

$$\Delta R \simeq \left.\frac{dR}{dX}\right|_{X_0} \Delta X. \tag{1.1}$$

Small changes in the readout are proportional to small changes in the target variable even when the full response curve is nonlinear. This local linearization is the ordinary logic of calibration.

The readout can also depend on nuisance variables. A spring may respond weakly to temperature or pressure. The height of mercury in a thermometer could respond to orientation through gravity, to properties of the glass, or to slow changes in the materials. A measurement protocol does not prove that these dependencies are absent; it holds them fixed or shows that they are negligible on the required scale.

> **Question from the audience.** Could the local gravitational field, or the direction in which the thermometer is held, change the reading?

Figure 1.1: A smooth but nonlinear thermometer response, with a short interval marked out for linear calibration.

Lecture frame: 00:07:55.

Response. Yes in principle. If orientation has an appreciable effect, the instruction to hold the thermometer vertically becomes part of the instrument's definition. Near Earth's surface the gravitational field is sufficiently stable that the remaining variation is normally negligible. [1]

Question from the audience. Could aging or chemical change slowly alter the calibration?

Response. It could. Mercury and glass are sufficiently inert for ordinary measurements, while changes on fantastically long time scales are irrelevant to the experiment. Calibration is always a statement about a specified instrument, range, protocol, and duration. [2]

For a Celsius thermometer, two reproducible reference states supply a convenient scale. If h_f and h_b are the column heights at the freezing and boiling points of water, respectively, then within a range where the response is nearly linear,

$$T_{\mathrm{C}} = 100\,\frac{h - h_f}{h_b - h_f}. \tag{1.2}$$

The choice $T_{\mathrm{C}} = 0$ at the freezing point is conventional. It is not the state of zero thermal energy, just as zero degrees Fahrenheit on a winter day is not absolute zero.

[1] Lecture timestamp: 00:04:08–00:05:00.
[2] Lecture timestamp: 00:05:03–00:06:11.

1.2 Equilibrium and Temperature as an Energy Scale

A thermometer measures the temperature of its surroundings only after the thermometer and the nearby medium have reached thermal equilibrium. Before then, its reading is merely changing toward equilibrium. The practical test is elementary: wait until the reading stops changing. The simplicity of that rule hides the physical processes that set the waiting time.

> **Question from the audience.** Does equilibration involve only a boundary layer around the bulb, and how long should one wait?
>
> **Response.** Energy must cross the glass and be exchanged with the surrounding medium, so the time depends on the thermometer, its thermal conductivity, and the medium. The immediate surroundings must also be representative of a uniform equilibrium state. Experimentally, the reading is accepted when it has ceased to drift. [3]

Temperature and density play different roles in this waiting time. A very dilute gas can be very hot: the few molecules present may carry large kinetic energies. Yet those molecules strike the thermometer rarely, so equilibration can be slow. Lowering density by a factor of one hundred or one thousand while holding temperature fixed does not lower the molecular energy scale; it lowers the rate at which that energy is communicated to the instrument.

> **Question from the audience.** Can the radiation in a room have a different temperature from the air?
>
> **Response.** Yes. Matter and radiation share one temperature only when they are in thermal equilibrium with one another. Neutral air couples weakly to much of the optical radiation in an illuminated room, and the photon population need not be an equilibrium blackbody field at the air temperature. [4]

Once equilibrium is established, temperature is best regarded as an energy scale. We continue to use units in which $k_B = 1$, so T itself has dimensions of energy. For a monatomic ideal gas,

$$E_0 = \frac{3}{2}NT. \tag{1.3}$$

The thermal energy per molecule is fixed by T; the corresponding speed still depends on the molecular mass.

1.3 Turning On a Weak Interaction

> **Question from the audience.** Under what conditions does the ideal-gas approximation break down, and can the breakdown be shown mathematically?
>
> **Response.** Start from the solved noninteracting gas and turn on a small interaction. Expand in that change, calculate the first correction, and compare it with the ideal term. The approximation fails when the correction is no longer small. [5]

[3] Lecture timestamp: 00:08:45–00:10:13.
[4] Lecture timestamp: 00:11:26–00:12:20.
[5] Lecture timestamp: 00:12:25–00:14:40.

This is the standard perturbative strategy. If a problem is soluble at $\lambda = 0$, introduce a coupling λ, write

$$F(\lambda) = F(0) + \lambda F_1 + \lambda^2 F_2 + \cdots \,, \tag{1.4}$$

and ask whether each retained term is smaller than the preceding one. Here the reference problem is the ideal gas. The physical smallness comes from a dilute sample, a short interaction range compared with the mean separation, and a potential weak enough for the first interaction term to be useful. High density or a long-range force can invalidate the approximation even when any one encounter seems mild.

Let $x_i \in \mathbb{R}^3$ and $p_i \in \mathbb{R}^3$ be the position and momentum of molecule i. For identical molecules of mass m, take

$$H(X, P) = \sum_{i=1}^{N} \frac{p_i^2}{2m} + \sum_{i<j} u(|x_i - x_j|)\,, \tag{1.5}$$

where $X = (x_1, \ldots, x_N)$ and $P = (p_1, \ldots, p_N)$. The restriction $i < j$ excludes self-pairs and counts (i, j) only once:

$$\sum_{i<j} 1 = \binom{N}{2} = \frac{N(N-1)}{2}. \tag{1.6}$$

Question from the audience. Why not sum independently over both particle labels?

Response. The pair $(3, 7)$ is the same physical pair as $(7, 3)$, and a particle is not paired with itself. Restricting the sum to $i < j$ enforces both facts. The mass symbol is kept as m, while i, j label particles, to avoid the notational collision that can occur when m is used for both purposes. [6]

Write the total configurational potential as

$$U(X) = \sum_{i<j} u(|x_i - x_j|)\,. \tag{1.7}$$

The interaction is assumed translationally invariant and short-ranged: $u(r)$ becomes negligible when r is large compared with a molecular length.

1.4 The Integrated Pair Strength

Only one integral of the weak pair potential enters the first correction:

$$U_0 \equiv \int_{\mathbb{R}^3} d^3r\, u(r). \tag{1.8}$$

Its units are energy times volume. It combines the typical strength of the interaction with the volume over which that interaction is appreciable.

To see why U_0 is useful, consider one pair:

$$I_2 = \int_V d^3x_1\, d^3x_2\, u(|x_1 - x_2|)\,. \tag{1.9}$$

[6] Lecture timestamp: 00:17:52–00:19:12.

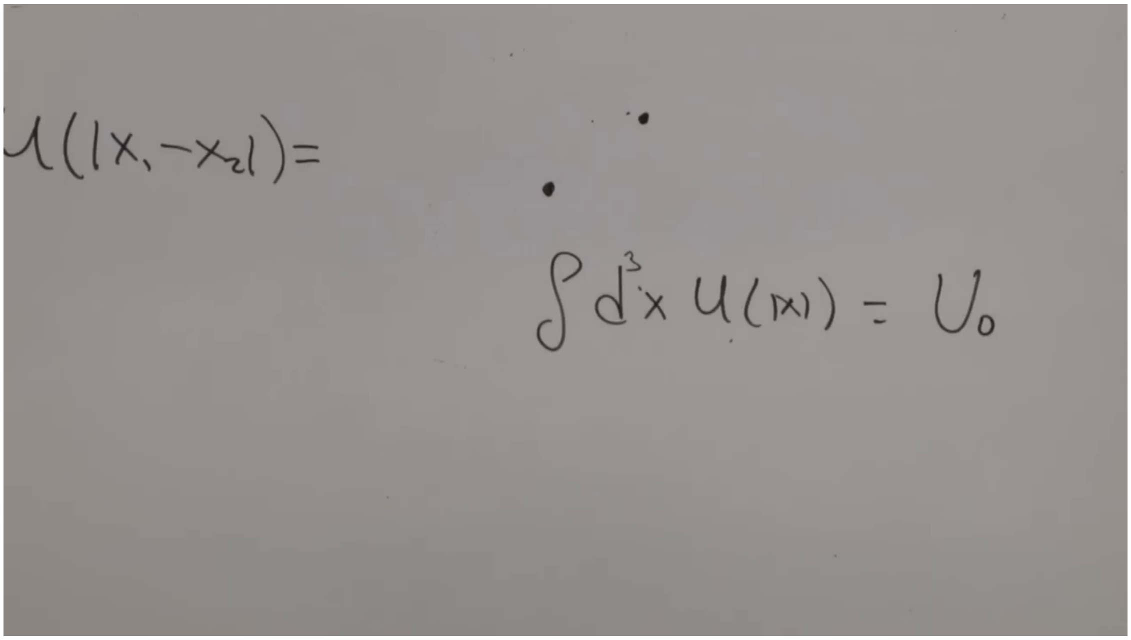

Figure 1.2: The short-range pair potential compressed into the integrated strength $U_0 = \int d^3r\, u(r)$.
Lecture frame: 00:25:15.

Introduce a relative coordinate $r = x_2 - x_1$. Away from the boundary, the relative integral gives U_0, while the common position of the pair ranges through the full container. Thus

$$I_2 = VU_0 + \text{surface corrections}. \tag{1.10}$$

The correction comes from particles within one interaction range of the wall, where part of the relative-coordinate neighborhood lies outside the box. Its fractional importance scales as a surface-to-volume ratio and vanishes in the macroscopic limit.

Question from the audience. Does U_0 refer to one selected pair, and is one particle kept at a fixed separation from the other?

Response. It refers to the common interaction law for any one pair. Hold the first particle fixed and integrate the second over every relative position. The separation is not fixed during this integral. Only the nearby interaction region contributes appreciably, producing U_0. [7]

Question from the audience. Why does the second position integral give the full container volume while the first does not?

Response. After the relative configuration has been integrated into U_0, the whole pair may be translated through the sample. That common translation supplies the large factor V. The interaction neighborhood is a molecular-scale volume already contained inside U_0, not another factor equal to the container volume. [8]

[7]Lecture timestamp: 00:26:03–00:27:33.
[8]Lecture timestamp: 00:27:48–00:29:21.

Question from the audience. Could the value depend on which point is held fixed, especially near a wall?

Response. Translational invariance makes the bulk value independent of the fixed point. Near a wall the integral is slightly different because some neighboring positions are unavailable. Only a surface layer of molecular thickness is affected, so the error is negligible when the box is macroscopic. [9]

Question from the audience. Is the result the potential energy integrated over every two-particle configuration, and why is it proportional to only one large volume?

Response. Yes. Its units are energy times volume squared, since VU_0 contains one container volume and one molecular interaction volume. Only the first of those grows with the size of the sample. [10]

Question from the audience. Does choosing a sphere, which minimizes area at fixed volume, change the bulk answer?

Response. It changes only the small boundary correction. For any ordinary macroscopic shape, the number of particles within a molecular distance of the surface is negligible compared with the number in the bulk. The surface becomes important only when the container itself approaches molecular scales. [11]

Question from the audience. Have collisions been omitted by replacing particle trajectories with a potential?

Response. No. The force is the derivative of the potential and is precisely what produces the collisions. Equilibrium statistical mechanics does not require us to follow those collisions one by one; their effect is encoded in the energy and therefore in the Boltzmann weight. [12]

Question from the audience. What is lost by retaining pair forces but not genuine three-body interactions?

Response. A microscopic energy can contain genuine three-body terms, and higher orders of a pair potential also generate many-particle correlations. In a dilute gas, however, three specified particles are much less likely than two to occupy one molecular neighborhood. Pair effects therefore give the first density correction; the higher clusters enter at higher powers of density. [13]

[9] Lecture timestamp: 00:29:24–00:31:22.
[10] Lecture timestamp: 00:31:31–00:32:04.
[11] Lecture timestamp: 00:32:04–00:33:58.
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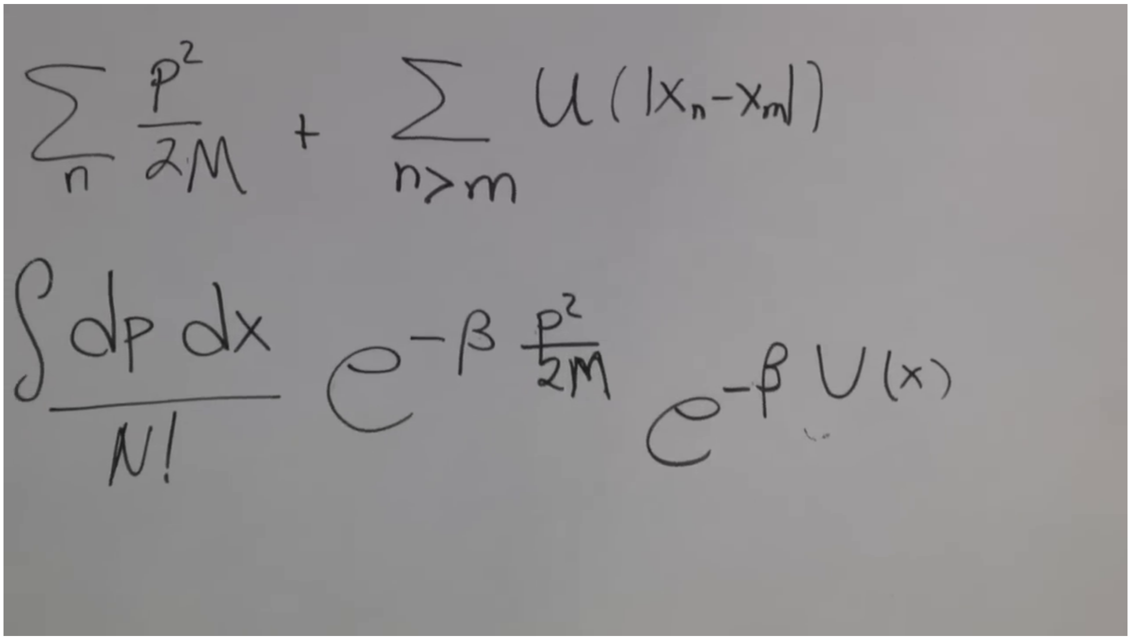

Figure 1.3: The many-particle energy and the corresponding kinetic and configurational factors in the partition function.

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1.5 The Partition Function on Autopilot

The advantage of the statistical method is now visible. Once the Hamiltonian is specified, there is no need to guess the detailed velocity distribution near walls or during collisions. We write the canonical partition function and follow its algebra:

$$\begin{aligned} Z &= \frac{1}{N!h^{3N}} \int d^{3N}P\, d^{3N}X \\ &\qquad \times e^{-\beta K(P)} e^{-\beta U(X)}, \\ K(P) &= \sum_i \frac{p_i^2}{2m}. \end{aligned} \tag{1.11}$$

The factor h^{3N} makes the classical phase-space measure dimensionless, and $N!$ removes the permutation overcounting of identical particles.

Momentum and position separate. Multiplying and dividing the configurational integral by V^N isolates the ideal-gas result:

$$\begin{aligned} Z &= \left[\frac{V^N}{N!h^{3N}} \int d^{3N}P\, e^{-\beta K(P)} \right] \\ &\quad \times \left[\frac{1}{V^N} \int d^{3N}X\, e^{-\beta U(X)} \right] \\ &\equiv Z_0\, C(\beta, V, N). \end{aligned} \tag{1.12}$$

The familiar Gaussian factor is

$$\int d^{3N}P\, e^{-\beta K(P)} = \left(\frac{2\pi m}{\beta} \right)^{3N/2}. \tag{1.13}$$

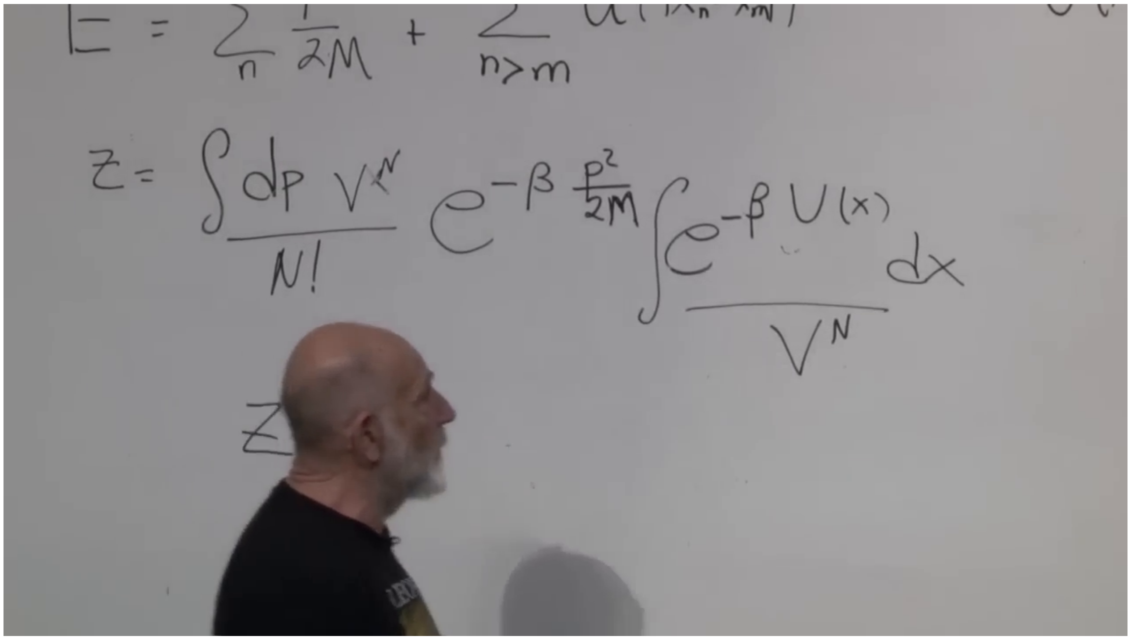

Figure 1.4: The ideal-gas partition function isolated from the normalized configurational factor.
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The $N!$ factor does not affect the normalized ratio C, nor the particular fixed-N derivatives used below, but it is not dispensable from the thermodynamics as a whole: it is essential for correct state counting and extensive entropy.

To first order in the interaction,

$$e^{-\beta U(X)} = 1 - \beta U(X) + O(U^2). \tag{1.14}$$

Consequently,

$$\begin{aligned} C &= 1 - \frac{\beta}{V^N} \sum_{i<j} \int d^{3N}X\, u(|x_i - x_j|) + O(U^2) \\ &= 1 - \frac{\beta}{V^N} \frac{N(N-1)}{2} V^{N-2}(VU_0) + O(U^2) \\ &= 1 - \beta \frac{N(N-1)}{2V} U_0 + O(U^2). \end{aligned} \tag{1.15}$$

Each pair has the same *integral*, not the same instantaneous separation or potential energy. The remaining $N-2$ position integrals each contribute one factor of V.

Question from the audience. Does the leading 1 assume that the potential vanishes at zero separation?

Response. No. It is the zeroth-order term in the exponential series, independent of the value of $u(0)$. The potential enters in the next term, $-\beta U(X)$. [14]

[14] Lecture timestamp: 00:46:19–00:46:48.

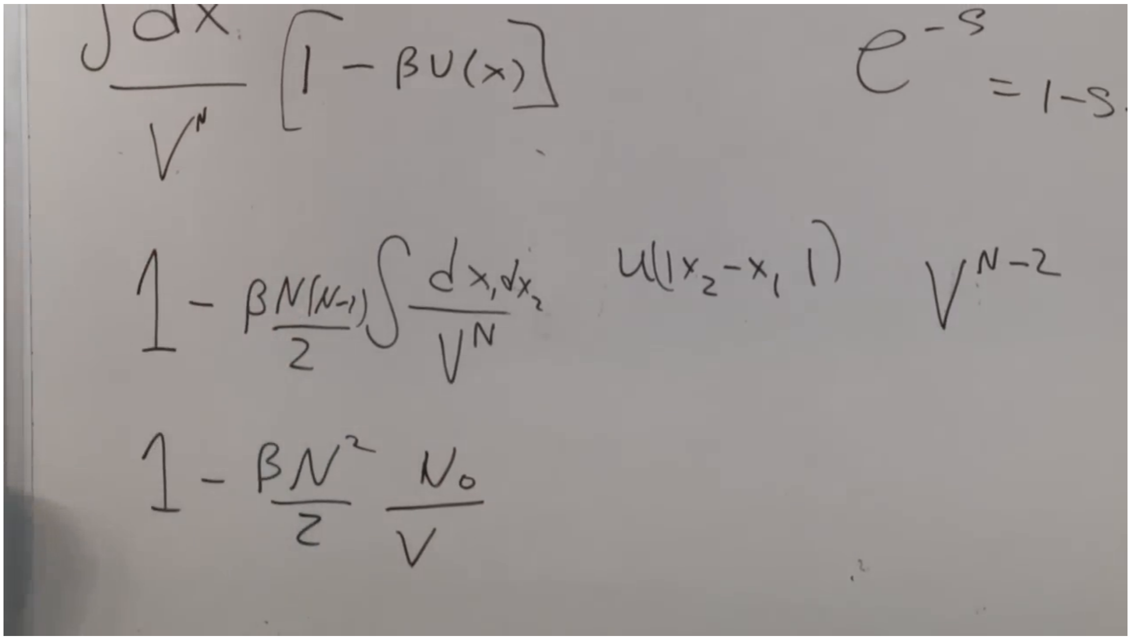

Figure 1.5: Pair counting, spectator-coordinate volumes, and the first configurational correction $1 - \beta N^2 U_0/(2V)$.

Lecture frame: 00:53:40.

> **Question from the audience.** Why may every pair be replaced by the same expression if different pairs have different separations?
>
> **Response.** Their pointwise values are not being equated. Every term is integrated over the same full configuration space, and particle labels do not distinguish one pair from another. The integrals have identical form and identical value. [15]

> **Question from the audience.** Is the number of pairs $N(N-1)/2$ or $N(N+1)/2$?
>
> **Response.** It is $N(N-1)/2$: after selecting one particle there are only $N-1$ distinct partners, and division by two removes the reversed ordering. For macroscopic N, replacing $N(N-1)$ by N^2 changes only a relative $1/N$ correction. [16]

Since $\log(1-x) = -x - x^2/2 - x^3/3 - \cdots$, not an alternating series, the first-order logarithm is

$$\log Z = \log Z_0 - \beta \frac{N(N-1)}{2V} U_0 + O(U^2). \tag{1.16}$$

At large N we may replace $N(N-1)$ by N^2 where no finite-N distinction is needed.

Remark 1.1. The controlled low-density formulation uses the Mayer function $f(r) = e^{-\beta u(r)} - 1$ and the second virial coefficient

$$B_2(T) = -\frac{1}{2}\int d^3r\, f(r). \tag{1.17}$$

For a weak potential, $f(r) \simeq -\beta u(r)$, so $B_2 \simeq \beta U_0/2$ and the result below is recovered. This form is needed for a hard repulsive core, where expanding $e^{-\beta u}$ directly is not valid.

[15] Lecture timestamp: 00:50:00–00:51:07.

[16] Lecture timestamp: 00:48:33–00:49:56.

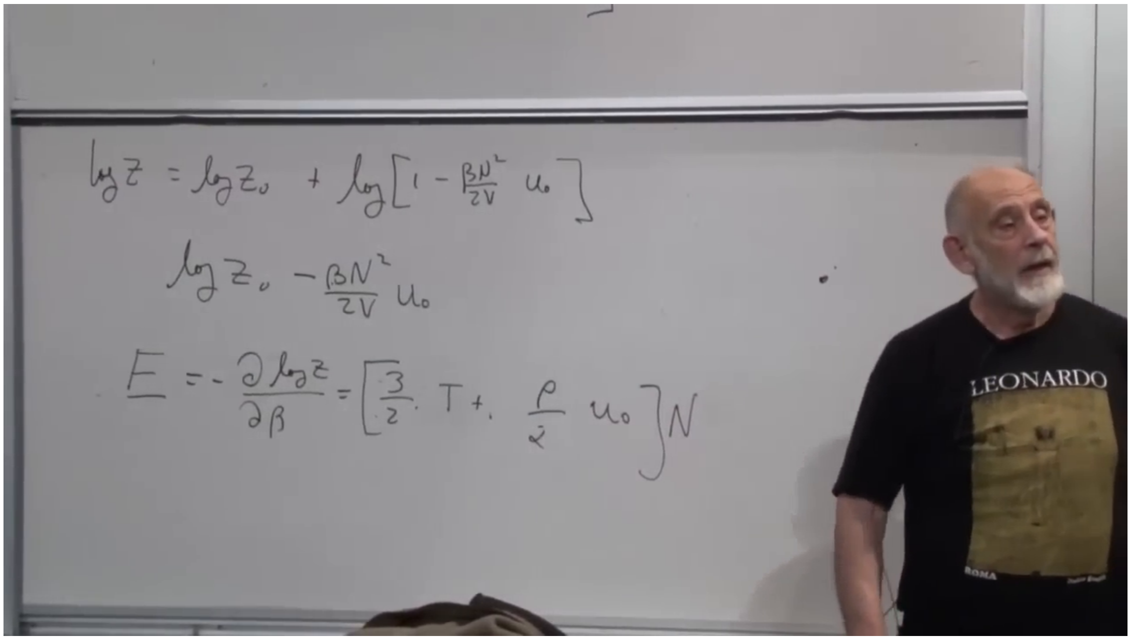

Figure 1.6: The first interaction correction reorganized as an extensive energy $N[3T/2 + \rho U_0/2]$.
Lecture frame: 01:03:25.

1.6 Energy, Pressure, and the Meaning of the Correction

The logarithm of Z is useful because its derivatives are measurable. Assuming the microscopic potential is independent of temperature,

$$\begin{aligned} E &= -\left(\frac{\partial \log Z}{\partial \beta}\right)_{V,N} \\ &= \frac{3}{2}NT + \frac{N(N-1)}{2V}U_0 + O(U^2) \\ &\simeq N\left(\frac{3}{2}T + \frac{1}{2}\rho U_0\right), \qquad \rho = \frac{N}{V}. \end{aligned} \tag{1.18}$$

The apparently nonextensive N^2 is accompanied by $1/V$. At fixed density, $N^2/V = N\rho$, so the total energy remains proportional to the size of the system. The extra energy per particle is proportional to density because a chosen molecule acquires pair energy only when another molecule lies nearby.

Pressure is easier to measure. Using the result of the preceding chapter,

$$P = T\left(\frac{\partial \log Z}{\partial V}\right)_{T,N}, \tag{1.19}$$

we obtain

$$\begin{aligned} P &= \frac{NT}{V} + T\frac{\partial}{\partial V}\left(-\beta\frac{N(N-1)}{2V}U_0\right) + O(U^2) \\ &= \rho T + \frac{1}{2}\rho^2 U_0 + O\left(\frac{\rho U_0}{N}, U^2\right). \end{aligned} \tag{1.20}$$

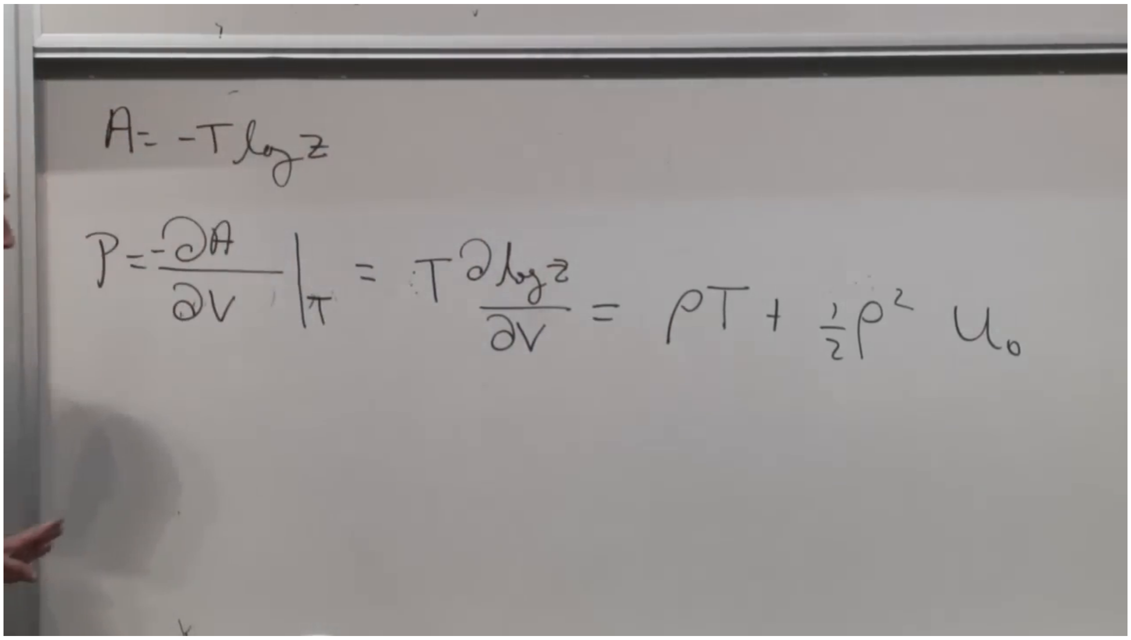

Figure 1.7: Pressure from the Helmholtz derivative, including the first quadratic-density correction.

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Question from the audience. Why is ρ^2, rather than another power, the natural first correction?

Response. The ideal term requires one particle to be present in a small region and therefore scales as ρ. An interaction requires two particles to occupy one molecular neighborhood, whose probability scales as ρ^2. The factor 1/2 is the same pair-counting factor that prevents (i, j) and (j, i) from being counted separately. [17]

At fixed temperature one can vary V, measure $\rho = N/V$, and plot the pressure. The ideal law is the linear term. Curvature of $P(\rho)$ reveals the first interaction correction. Within this weak-potential approximation, the correction has no explicit temperature dependence because the factor T in (1.19) cancels $\beta = 1/T$.

Question from the audience. How large may the density become before three-body effects must be included?

Response. Density counting says that the next virial term is proportional to ρ^3, but its coefficient cannot be inferred from U_0 alone. It receives contributions from correlated products of pair interactions and, if present, genuine three-body forces. One must calculate the next cluster coefficient and compare it with the ρ^2 term; a guessed formula should not be promoted to a result. [18]

The ideal-gas term dominates when the magnitude of the correction is small:

$$\begin{aligned} \frac{1}{2}\rho^2|U_0| &\ll \rho T, \\ |\rho U_0| &\ll T \qquad \text{(parametrically).} \end{aligned} \tag{1.21}$$

[17] Lecture timestamp: 01:24:27–01:26:00.
[18] Lecture timestamp: 01:10:50–01:13:38.

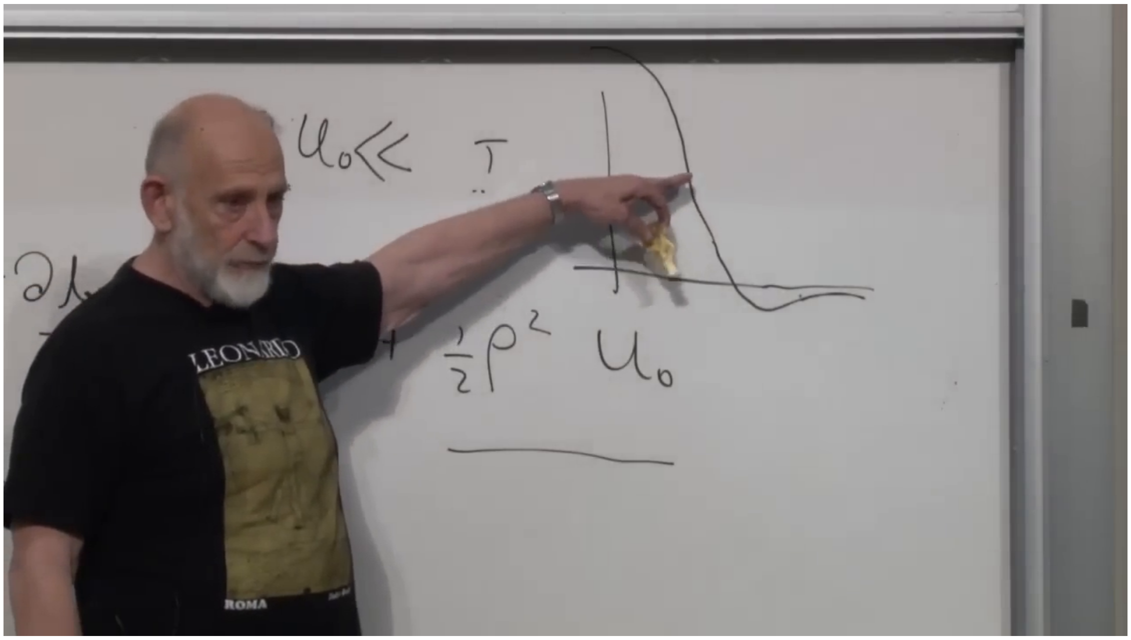

Figure 1.8: Qualitative repulsive and attractive pair-potential profiles beside the pressure formula whose correction follows the sign of U_0.

Lecture frame: 01:21:30.

Both sides of the second inequality are energies. The quantity ρU_0 is the interaction-energy scale per particle, while T is the kinetic-energy scale. An ordinary dilute room-temperature gas lies far inside this regime; substantial deviations arise only when molecular neighborhoods are occupied often enough for interactions to compete with thermal motion.

Question from the audience. Can the criterion be estimated for the air in this room?

Response. Estimate U_0 as a molecular interaction energy times a molecular volume, multiply by the number density, and compare the result with $k_B T$. The rough molecular numbers show that the interaction scale is much smaller in ordinary room air. Considerable compression is required before molecules frequently crowd into one another's interaction range. [19]

Question from the audience. May U_0 have either sign?

Response. Yes. A positive integrated interaction is net repulsive in this approximation and raises the pressure above ρT. A negative one is net attractive and lowers it. This sign test is an immediate physical check on (1.20). [20]

Question from the audience. Can electrostatic repulsion, gravitational attraction, or real molecular forces be inserted into this calculation?

[19] Lecture timestamp: 01:16:24–01:18:54.
[20] Lecture timestamp: 01:18:56–01:20:29.

Response. The microscopic potential is input from atomic, molecular, or quantum physics. Neutral molecules commonly exhibit a steep short-range repulsive core and a longer attractive van der Waals tail, often represented schematically by a 12–6 potential. The full Mayer-function formula should be used for such a hard core. Gravity is different: it is long-ranged, its spatial integral does not converge in the required way, and the ordinary short-range thermodynamic limit used here fails. [21]

The calculation illustrates a useful division of labor. Microscopic physics supplies $u(r)$. Statistical mechanics turns that energy into Z, derivatives of Z produce bulk observables, and physical reasoning checks their sign, density scaling, extensivity, and domain of validity. The formal manipulation is mechanical only after the correct energy and approximation have been chosen.

1.7 Exact Differentials and Path Dependence

We now change mathematical language. In the phrase *exact differential*, exact does not mean numerically precise. It means that a one-form is the differential of a genuine scalar function.

Let $f(x, y)$ be the altitude above a point in a landscape. A small displacement changes it by

$$df = \frac{\partial f}{\partial x}\, dx + \frac{\partial f}{\partial y}\, dy = f_x\, dx + f_y\, dy. \tag{1.22}$$

Suppose instead that we are merely handed

$$\omega = F_x(x, y)\, dx + F_y(x, y)\, dy. \tag{1.23}$$

Does a scalar f exist with $\omega = df$? Equality of mixed partials requires

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}. \tag{1.24}$$

On a simply connected domain this zero-curl condition is also sufficient. The components (F_x, F_y) then form the gradient of f.

The same condition has an integral meaning. If $\omega = df$, then

$$\int_{\gamma:1\to 2} \omega = f(2) - f(1) \tag{1.25}$$

depends only on the endpoints, not on the path γ. In particular,

$$\oint_{\gamma} \omega = 0 \tag{1.26}$$

for every contractible closed loop.

Altitude has this property. Driving between the same two points by a level road or over a hill gives the same net change in altitude. Fuel use does not. A hilly route and a level route generally consume different amounts, and a closed trip does not restore the fuel that was burned. The amount of fuel remaining is certainly a well-defined quantity in an enlarged state description, but it is not a function of the car's position (x, y) alone; it carries memory of the route. Tiredness supplies the same kind of analogy: descending a hill does not generally undo the fatigue of climbing it.

[21] Lecture timestamp: 01:20:40–01:23:03.

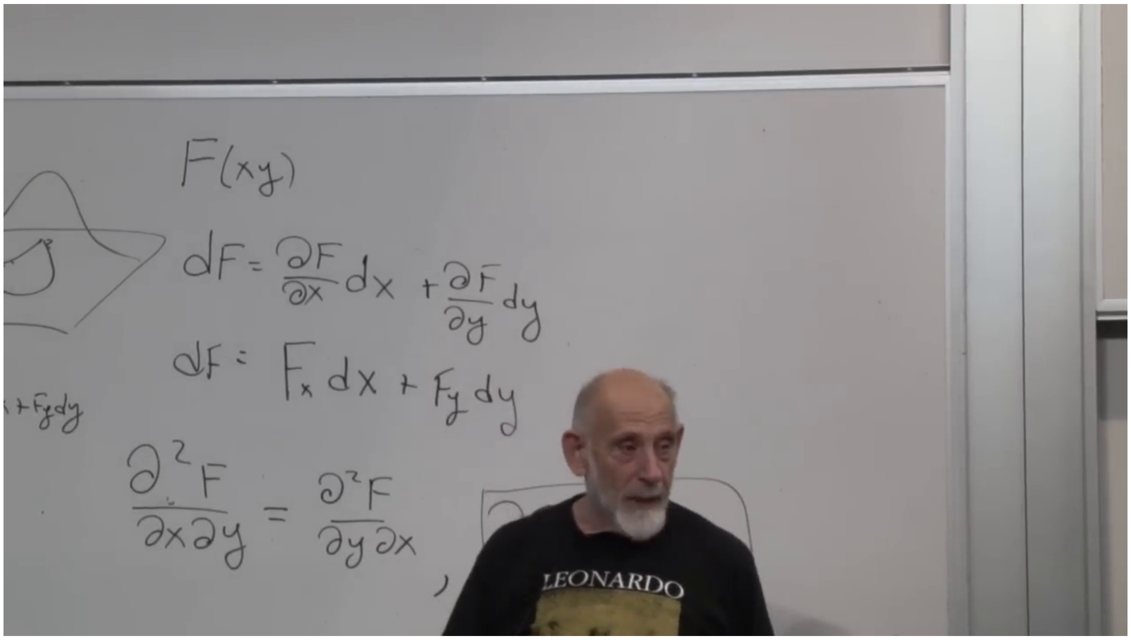

Figure 1.9: Altitude as a scalar function, its differential, and the mixed-partial condition for exactness.

Lecture frame: 01:34:30.

Question from the audience. Is exactness the same idea as a conservative force, and is it related to curvature?

Response. A force field with zero curl is locally conservative and admits a potential, so the language is closely related. Parallel transport and curvature also connect closed-loop changes to local geometric data, but that is a broader subject. Here only the elementary statement is needed: zero curl means local path independence, with the usual global condition on the domain. [22]

Two examples make the test concrete. First take

$$F_x = y, \qquad F_y = x. \tag{1.27}$$

Both cross derivatives equal 1, and indeed

$$\omega = y\,dx + x\,dy = d(xy). \tag{1.28}$$

Now change one sign:

$$F_x = y, \qquad F_y = -x. \tag{1.29}$$

Then

$$\frac{\partial F_x}{\partial y} = 1, \qquad \frac{\partial F_y}{\partial x} = -1, \tag{1.30}$$

so no scalar $f(x, y)$ has these two components as its gradient.

[22] Lecture timestamp: 01:39:22–01:41:07.

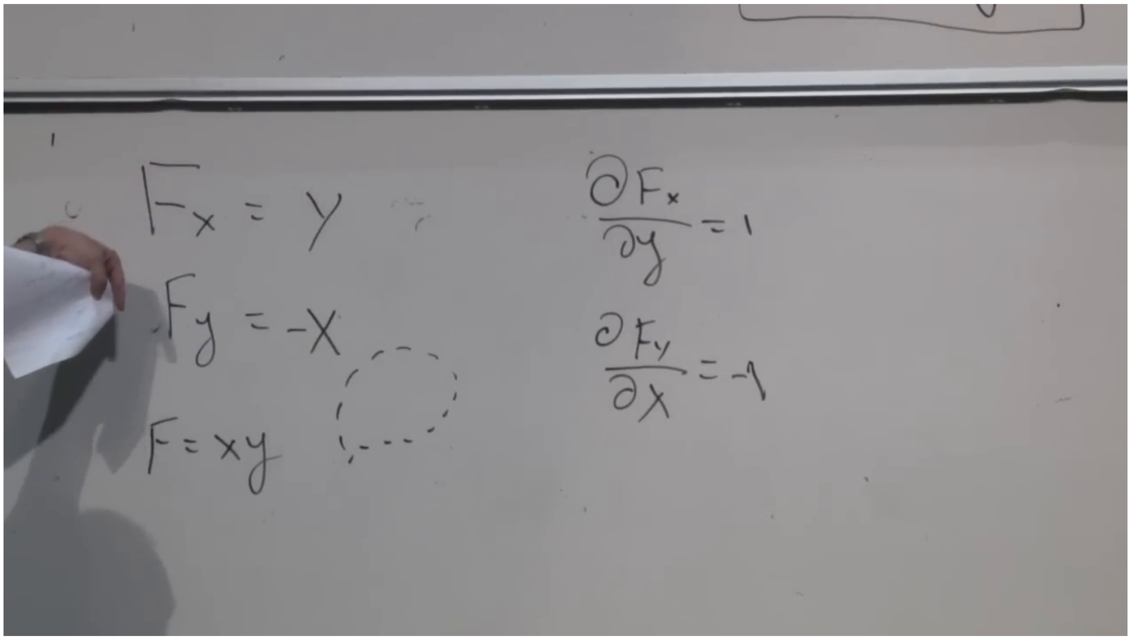

Figure 1.10: An exact example generated by $f = xy$ and an inexact example whose cross derivatives are 1 and -1.

Lecture frame: 01:44:50.

1.8 Heat and Work Are Not State Functions

For a simple compressible system with fixed particle number, two independent equilibrium variables determine the state. We may use (T, V), (E, V), or, especially naturally for the fundamental relation, (S, V). Thus

$$E = E(S, V), \tag{1.31}$$

and its exact differential is

$$dE = \left(\frac{\partial E}{\partial S}\right)_V dS + \left(\frac{\partial E}{\partial V}\right)_S dV = T\, dS - P\, dV. \tag{1.32}$$

Question from the audience. Does adiabatic mean slow, insulated, reversible, or constant entropy?

Response. The terminology is used differently in different contexts. For the equilibrium path considered here, the change is quasistatic and thermally insulated, hence reversible and isentropic. A merely insulated irreversible process can still produce entropy, while the quantum adiabatic theorem emphasizes slow parameter variation. [23]

At fixed entropy, an expansion changes the gas energy by $-P\, dV$. At fixed volume, a reversible transfer of heat changes it by $T\, dS$. It is therefore useful to write

$$\begin{aligned} \delta W_{\text{on}} &= -P\, dV, \qquad \delta Q_{\text{rev}} = T\, dS, \\ dE &= \delta Q_{\text{rev}} + \delta W_{\text{on}}. \end{aligned} \tag{1.33}$$

[23] Lecture timestamp: 01:47:13–01:47:54.

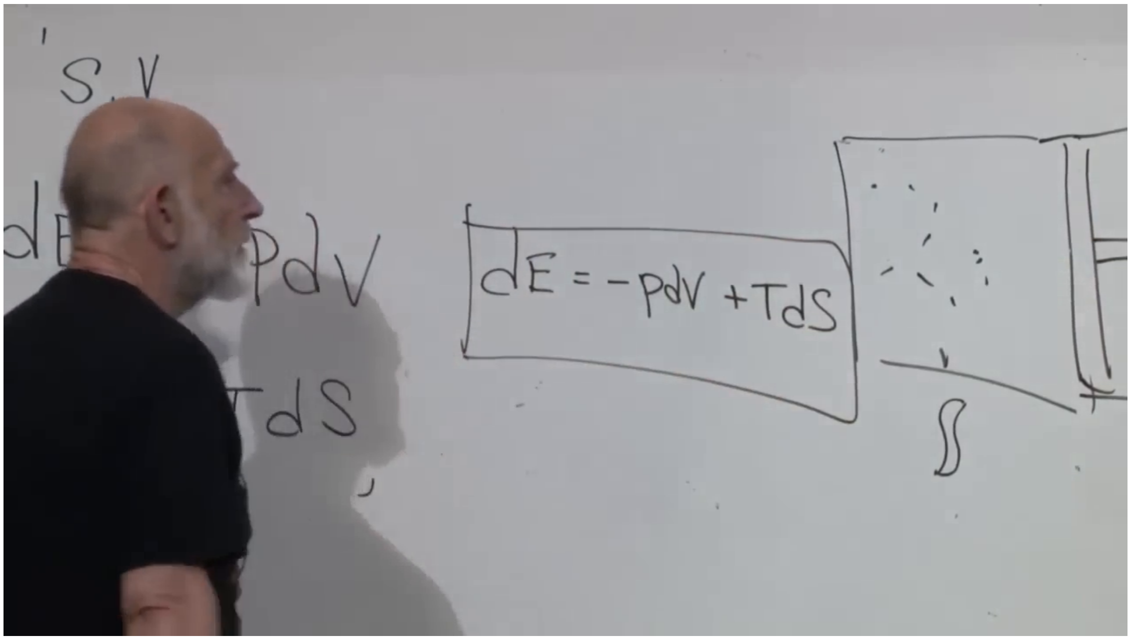

Figure 1.11: The first law for a piston system: $dE = -P\,dV + T\,dS$.
Lecture frame: 01:50:28.

The symbol δ, rather than d, anticipates that heat and work are path increments, not differentials of state functions.

> **Question from the audience.** Does $-P\,dV$ represent work done on the gas or by the gas?
>
> **Response.** With the convention in (1.33), it is work done on the gas. During expansion $dV > 0$, the gas does positive work on the piston, so the work done on the gas is negative and the gas loses energy. The two descriptions are equal and opposite. [24]

The equality $\delta Q_{\mathrm{rev}} = T\,dS$ must not be extended unchanged to arbitrary irreversible heating. In general,

$$dS = \frac{\delta Q}{T_{\mathrm{boundary}}} + dS_{\mathrm{prod}}, \qquad dS_{\mathrm{prod}} \geq 0. \tag{1.34}$$

The exact-differential argument below concerns quasistatic equilibrium paths, for which the reversible relation is the appropriate one.

> **Question from the audience.** Are all paths in the state-space argument assumed to pass through equilibrium states?
>
> **Response.** Yes. The operations are performed slowly enough that P, T, E, S, and V remain well-defined along the path. The path dependence of heat and work does not require the system to be driven far from equilibrium. [25]

[24] Lecture timestamp: 01:51:23–01:52:27.
[25] Lecture timestamp: 01:54:30–01:54:39.

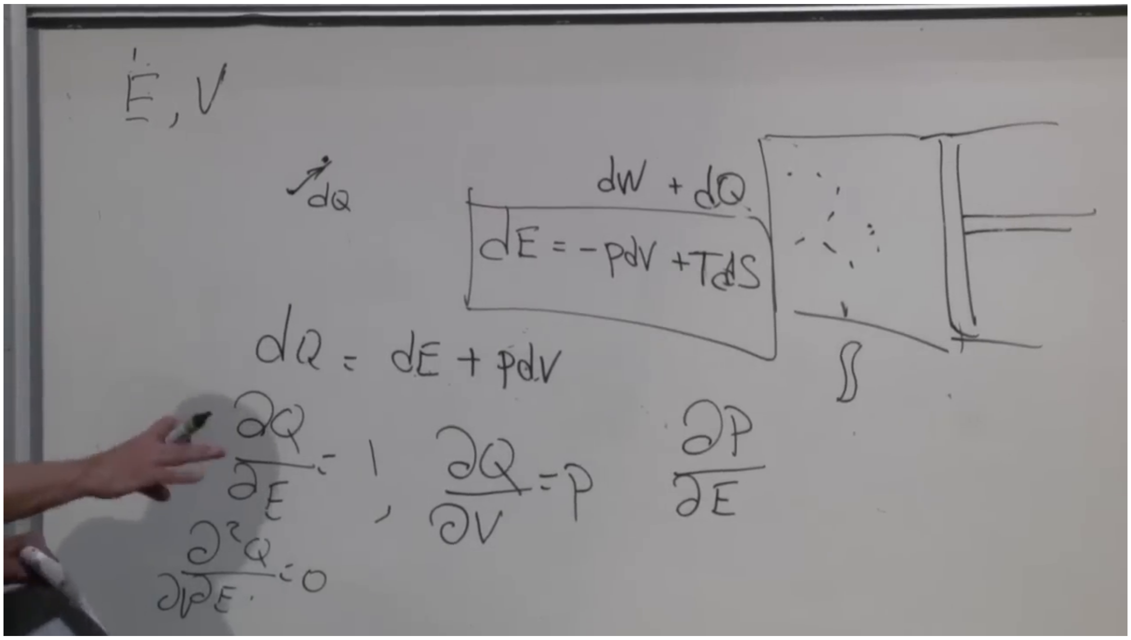

Figure 1.12: The heat one-form $dE + P\,dV$ and the mixed derivatives required if it were the differential of a state function.

Lecture frame: 01:58:10.

To test heat for exactness, use (E, V) as coordinates and rearrange the first law:

$$\delta Q_{\text{rev}} = dE + P(E, V)\,dV. \tag{1.35}$$

If there were a state function $Q(E, V)$ with $dQ = \delta Q_{\text{rev}}$, its partial derivatives would have to be

$$\left(\frac{\partial Q}{\partial E}\right)_V = 1, \qquad \left(\frac{\partial Q}{\partial V}\right)_E = P. \tag{1.36}$$

Equality of mixed derivatives would then require

$$\begin{aligned} \frac{\partial}{\partial V}\left(\frac{\partial Q}{\partial E}\right)_V &= \frac{\partial}{\partial E}\left(\frac{\partial Q}{\partial V}\right)_E, \\ 0 &= \left(\frac{\partial P}{\partial E}\right)_V. \end{aligned} \tag{1.37}$$

Question from the audience. Is this precisely the same curl test used for $F_x\,dx + F_y\,dy$?

Response. Yes. The coefficients of dE and dV are 1 and $P(E, V)$. Their cross derivatives are 0 and $(\partial P/\partial E)_V$. Heat would be exact only if those derivatives agreed. [26]

The requirement is already physically implausible: adding energy to a gas at fixed volume generally changes its pressure. The ideal gas supplies a direct calculation:

$$PV = NT, \qquad E = \frac{3}{2}NT \quad \Longrightarrow \quad P = \frac{2E}{3V}. \tag{1.38}$$

[26] Lecture timestamp: 01:54:39–01:54:52.

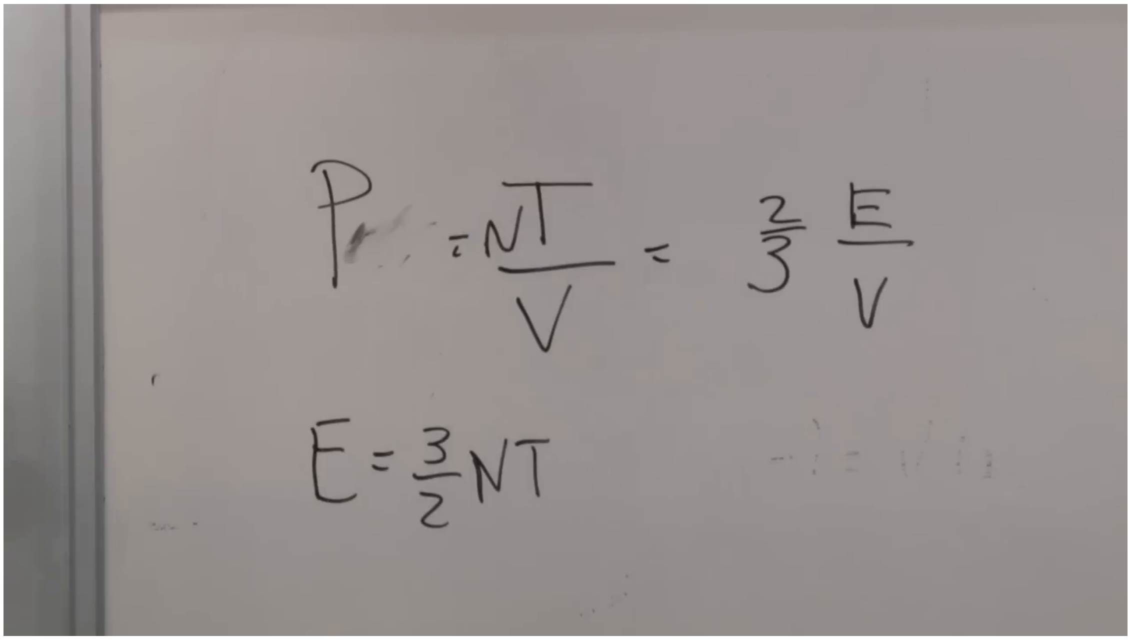

Figure 1.13: The ideal-gas relations combined into $P = 2E/(3V)$.

Lecture frame: 02:00:00.

Therefore

$$\left(\frac{\partial P}{\partial E}\right)_V = \frac{2}{3V} \neq 0. \tag{1.39}$$

The curl test fails: δQ_{rev} is not exact. The work increment $\delta W_{\mathrm{on}} = -P\,dV$ fails for the same reason.

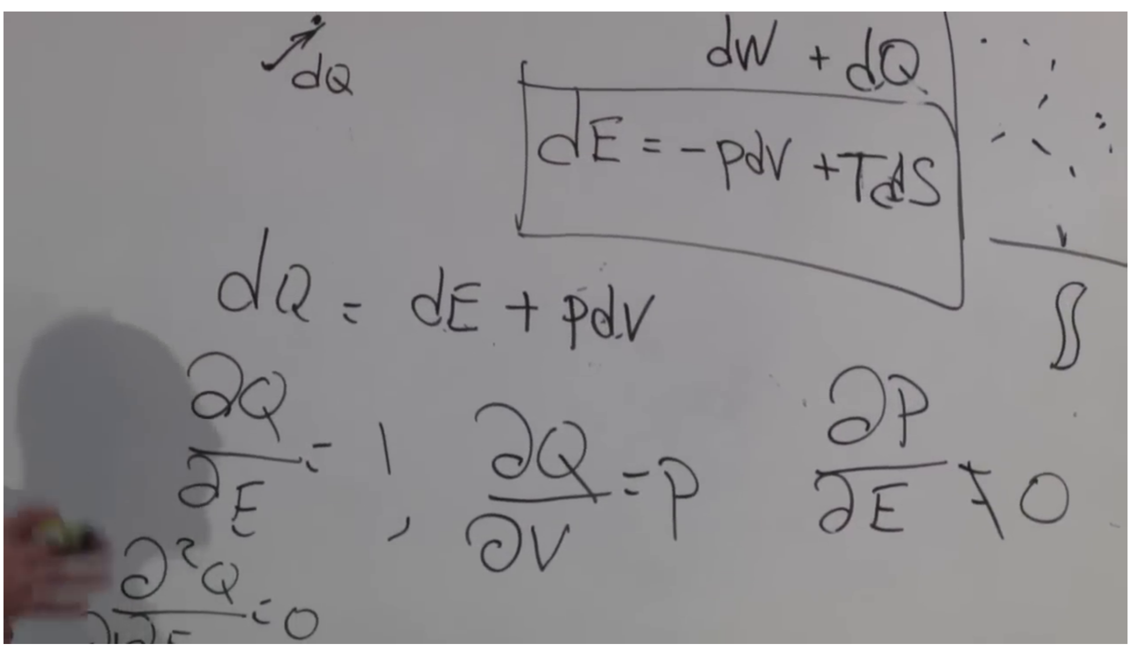

Figure 1.14: The completed argument: because $(\partial P/\partial E)_V \neq 0$, the heat one-form cannot be exact.

Lecture frame: 02:00:35.

For a closed equilibrium cycle,

$$\oint dE = 0, \qquad \oint \delta Q_{\text{rev}} = -\oint \delta W_{\text{on}}. \tag{1.40}$$

The system returns to the same energy, but heat and work need not vanish separately. Their values depend on the route through state space. This is why a cyclic engine can absorb heat and deliver net work even though its working substance returns to its initial thermodynamic state.

> **Question from the audience.** Is this path dependence connected with engine efficiency, hysteresis, and memory?
>
> **Response.** Engine cycles use it directly: the closed-cycle heat input balances the net work output while the state function E returns to its initial value. Hysteresis and memory also involve path dependence, although they often require additional internal variables or nonequilibrium physics. The present conclusion is narrower and already holds for reversible equilibrium paths: heat and work are transfers, not properties stored in the thermodynamic state. [27]

The two halves of the argument now meet. Measurement is meaningful only after its controls and equilibrium conditions are specified. Approximation is meaningful only after its small parameter and failure criterion are specified. A thermodynamic quantity is a state function only after its differential passes the path-independence test. In each case, disciplined mathematics does not replace physical interpretation; it tells us exactly where that interpretation is justified.

[27] Lecture timestamp: 02:02:21–02:03:17.